Proteoform Profiles Distinguish Adult Zebrafish Telencephalon and Optic Tectum

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Abstract

Adult zebrafish telencephalon and optic tectum stand on opposite sides of the forebrain-visuomotor divide, yet it has remained unclear how sharply that difference appears at the proteoform level. We analyzed the region-resolved top-down proteomics dataset of Lubeckyj and Sun (2022) together with the Tel2 and Teo2 proteoform spreadsheets deposited in PRIDE. The separation is sharp. Of the 807 proteoforms reported in the article, only 35 were shared between the two regions, and the same low-overlap pattern persisted after exact identifier matching, protein collapse, and conservative sensitivity analyses for uneven recovery and identifier error. The biology of the split was equally clear. Telencephalon was enriched for neuropeptide and synaptic proteoforms, whereas optic tectum was enriched for reticulon-, actin-, myosin-, and troponin-linked entries, and an independent literature-derived marker panel supported the same regional axis. The apparent excess of N-terminal acetylation in telencephalon weakened after correction for differential detectability. The deposited data therefore show that adult zebrafish telencephalon and optic tectum are organized by distinct proteoform profiles that mirror their different biological roles.

1 Introduction

Adult zebrafish telencephalon and optic tectum do very different work. The telencephalon participates in social behavior, learning, memory, and other forebrain functions (Stednitz et al., 2018; Reemst et al., 2023; Pandey et al., 2023). The optic tectum, by contrast, is a visuomotor center that transforms sensory input into orienting and avoidance behavior (Roeser and Baier, 2003; Orger, 2016). If that functional division is written into the protein state of each region, it should be visible most clearly at the proteoform level, where sequence variants and post-translational modifications are resolved rather than folded into protein totals.

The LCM-CZE-MS/MS study of Lubeckyj and Sun (2022) offers an unusual opportunity to ask that question directly. The article reported hundreds of proteoforms from adult zebrafish telencephalon and optic tectum and linked those profiles to regional function. Just as important, the study deposited two region-specific spreadsheets, `Proteoform_ID_Tel2.xlsx` and `proteoform_ID_Teo2.xlsx`, in the PRIDE proteomics repository. Those files expose the overlap structure of the dataset in a way that the narrative of the paper alone cannot. They allow direct comparison of exact accession+proteoform identities, duplicate incidence, protein-level collapse, and biologically interpretable marker families from the deposited evidence itself.

Read in that form, the dataset makes a sharper statement than the original article drew explicitly. Telencephalon and optic tectum share very few proteoforms, and the proteoforms that distinguish them fall along a biologically intelligible axis: neuropeptide and synaptic entries on the telencephalic side, reticulon- and cytoskeleton-linked entries on the tectal side. The apparent excess of N-terminal acetylation in telencephalon is much less secure and weakens once uneven recovery is taken into account. The analysis presented here was designed around that distinction. It asks how strong the regional separation remains under exact matching and conservative sensitivity analysis, whether the separating proteoforms align with known regional biology, and where the evidence becomes too thin to support a firm claim.

This question matters for more than one zebrafish dataset. Prior functional studies and atlases establish the broad biology of the two regions (Stednitz et al., 2018; Pandey et al., 2023; Roeser and Baier, 2003; Orger, 2016), and recent top-down work shows that proteoform-level structure in zebrafish brain remains informative well beyond a single pilot study (Askarani et al., 2025). At the same time, region-resolved datasets are often discussed at the level of appealing examples rather than explicit overlap structure. The present study shows that the deposited proteoform evidence is already strong enough to support a definite statement about regional molecular organization in the adult zebrafish brain.

2 Data and Analysis Setup

2.1 Source material

The analysis is built on Lubeckyj and Sun (2022), a pilot study of adult zebrafish telencephalon and optic tectum using laser capture microdissection coupled to capillary zone electrophoresis tandem mass spectrometry. Three public sources were used: the article itself, the journal supplementary workbook, and the two region-specific proteoform spreadsheets deposited in PRIDE, `Proteoform_ID_Tel2.xlsx` and `Proteoform_ID_Teo2.xlsx`. The supplement reports proteoform identification with TopPIC at 1% proteoform-spectrum-match FDR and 5% proteoform-level FDR, with duplicate label-free quantification in TopDiff. The `Tel2` and `Teo2` sheets embedded in the supplement match the PRIDE deposit exactly, so the supplement serves as provenance metadata and the deposited spreadsheets serve as the quantitative source. We parsed those spreadsheets directly, preserved row-level provenance in local CSV and JSON traceability files, and recorded the public raw-file links in the manifest. Full raw-file reprocessing was not attempted because the processed feature histories needed for a faithful rerun are not part of the deposit. The exact marker-to-row mappings used in the paper are materialized in `results/marker_family_membership.csv`; claim-to-file provenance is mirrored in `data/evidence_traceability.json`, and source URLs are listed in `data/source_manifest.json`.

For exact-ID overlap we treat the biological unit as the accession-plus-proteoform pair. This is stricter than comparing proteoform strings alone, because the same peptide-like string can appear under more than one accession in the deposited spreadsheets. When that occurred, accession identity was retained in the primary overlap analysis and collapse to broader protein units was treated only as a sensitivity check. For the curated marker panel, we use transparent matching rules tied to the biological language of the original paper: exact gene-symbol matches for NPY, PENKB, PYY-A, and neurogranin a, and description-keyword matches for synucleins, reticulon, myosin, actin, and troponin. Those rules are written into the local traceability files so that a reader can inspect or challenge them directly.

2.2 Derived metrics

The analysis focused on three features of the deposited evidence: the degree of regional separation, the biological direction of the separating proteoforms, and the stability of the apparent N-terminal acetylation difference. Regional separation was assessed with article-level counts, exact accession+proteoform overlap, canonicalized discrepancy diagnostics, protein-level collapse, abundance-weighted similarities, prevalence-adjusted overlap tests, a duplicate-based detectability correction, and conservative misidentification bounds keyed to the published FDR values. Biological direction was assessed with two panels: the marker families emphasized in the original paper and an independent literature-derived composition panel. The acetylation analysis was restricted to the matched marker subsets and then stress-tested against simple proxies for unequal recovery.

Additional robustness checks were limited to what the deposited spreadsheets can genuinely support: alternative abundance normalizations, duplicate-informed missingness bounds, a minimal MNAR-style low-intensity fill, family-size-preserving randomization, conservative spillover reassignment, motor-family exclusion, and protein collapse. For the reviewer-requested acetylation follow-up, we used a coarse logistic sensitivity model with region as the outcome and acetylation, log precursor mass, sequence span length, and first-residue position as predictors, followed by a restricted analysis on rows whose first residue lay at or within two amino acids of the N terminus. The formulas, normalization definitions, canonicalization rules, detectability model, and missingness calculations are given in the Appendix so that the main text can remain centered on the biological result.

2.3 Interpretive scope

The available material is sufficient for overlap analysis, marker-family mapping, and conservative PTM sensitivity checks, but not for a full raw-file reprocessing or feature-level error model. The strongest claims of the paper are therefore about regional separation and about the biological direction of that separation. Within those limits, the deposited spreadsheets are strong enough to support a definite statement about proteoform organization in the adult zebrafish brain.

3 Results

3.1 Telencephalon and optic tectum are sharply separated at the proteoform level

At the proteoform level, the two regions are far apart. The article reported 309 telencephalic proteoforms and 533 tectal proteoforms, with only 35 shared between them. Exact accession+proteoform matching of the deposited Tel2 and Teo2 spreadsheets is even stricter, yielding 29 shared entries among 805 total exact IDs. Protein-level overlap is broader, but still modest. By every measure used here, telencephalon and optic tectum do not look like lightly shifted versions of the same local proteoform pool.

That conclusion does not depend on one encoding choice. Abundance weighting changes the overlap only slightly, and prevalence-adjusted testing places the observed exact-ID overlap far below the fixed-margin independence baseline (expected Jaccard 0.3152, centered Jaccard -0.2792, $p = 5.56 \times 10^{-183}$). The difference between the article's 35 shared proteoforms and the spreadsheet-derived exact-ID count of 29 appears to be representational rather than biological: relaxed matching that strips bracketed PTM annotations and punctuation while keeping accessions fixed raises the shared count to 44, but the overlap remains very low even in that view (canonicalized Jaccard 0.0615, centered Jaccard -0.2627, $p = 5.01 \times 10^{-143}$). The biological picture is the same in every case.

Sensitivity analysis leads to the same end point. Duplicate-informed missingness bounds, jackknife estimates, and minimal MNAR-style fills never turn the dataset into a high-overlap case. In a simple occupancy model built from duplicate rediscovery, correcting for uneven recovery raises the exact-ID Jaccard only from 0.0360 to 0.0432. Conservative identifier-error bounds are equally modest: even if 5% of region-unique exact IDs are treated as possible false discoveries in the direction most favorable to overlap, the Jaccard ceiling rises only to 0.0540. Uneven detectability and plausible identifier error can soften the edge of the split, but they do not erase it.

3.2 Marker families follow known regional biology

Low overlap by itself could still reflect noise. It does not. The curated marker panel is concentrated almost entirely in the region where those proteins would be expected from known biology. Telencephalon-associated markers contribute 23 matched counts and only 2 spillover counts, whereas optic-tectum-associated markers contribute 102 matched counts and 1 spillover count. Across the full panel, 125 of 128 observations fall in

Table 1: Core regional separation metrics supporting the regional-organization claim. The first two rows preserve the article-level summary, and the remaining rows use the deposited region spreadsheets plus detectability and identifier-error bounds.

Metric	Value
Article shared proteoforms	35
Article Jaccard overlap	0.0434
Exact-ID shared proteoforms	29
Exact-ID Jaccard overlap	0.0360
Weighted Jaccard overlap	0.0427
Protein overlap fraction	0.2151
Occupancy-adjusted shared exact IDs	43.2
Occupancy-adjusted Jaccard overlap	0.0432
5% misidentification ceiling	0.0540

Table 2: Curated marker families used in the axis-alignment analysis. Exact accession+proteoform members for each family are listed in the local marker-membership file, but the biological interpretation still follows the marker groups named in the original paper.

Marker group	Tel	Teo	Interpretation
Pro-neuropeptide Y	1	0	feeding-related neuropeptide signaling
Synenkephalin	1	0	affective signaling
Peptide YY-A	1	0	peptide hormone signaling
Neurogranin a	6	0	memory-linked synaptic signaling
Synucleins total	14	2	synaptic and mnemonic family
Reticulon	0	4	retinotectal organization
Myosin	0	48	motor machinery
Actin	0	19	cytoskeletal movement
Troponin	1	31	contraction-linked regulation

the expected region. Because optic tectum contributes the larger raw proteoform total in the original study, this alignment cannot be dismissed as a trivial consequence of abundance imbalance.

The statistical support is correspondingly strong. The panel yields an odds ratio of 1173 with an approximate 95% confidence interval from 102.0 to 13,495.1 and a one-sided exact p value of 5.12×10^{-22}. A family-size-preserving randomization reaches the same conclusion without relying on asymptotic logic: under 200,000 randomizations that preserved the nine family sizes and the global 25-versus-103 telencephalon/optic-tectum label totals, none matched the observed 125 correctly placed counts. All nine curated marker groups point in the expected direction, and the signal remains strong after protein collapse and intensity weighting. The separating proteoforms are therefore not random residues of sparse sampling. They trace a coherent biological axis.

The same pattern survives more stringent perturbations. Leave-one-out removal of any single marker family keeps the matched-region fraction between 0.9625 and 0.9911. Conservative spillover reassignment still leaves the panel strongly aligned, and removing the motor-associated myosin, actin, and troponin families leaves an alignment fraction of 0.9310. A simple contamination screen argues against an obvious purity artifact: the deposited spreadsheets contain only two optic-tectum glial-support sentinels but 23 telencephalon myelin sentinels versus three in optic tectum.

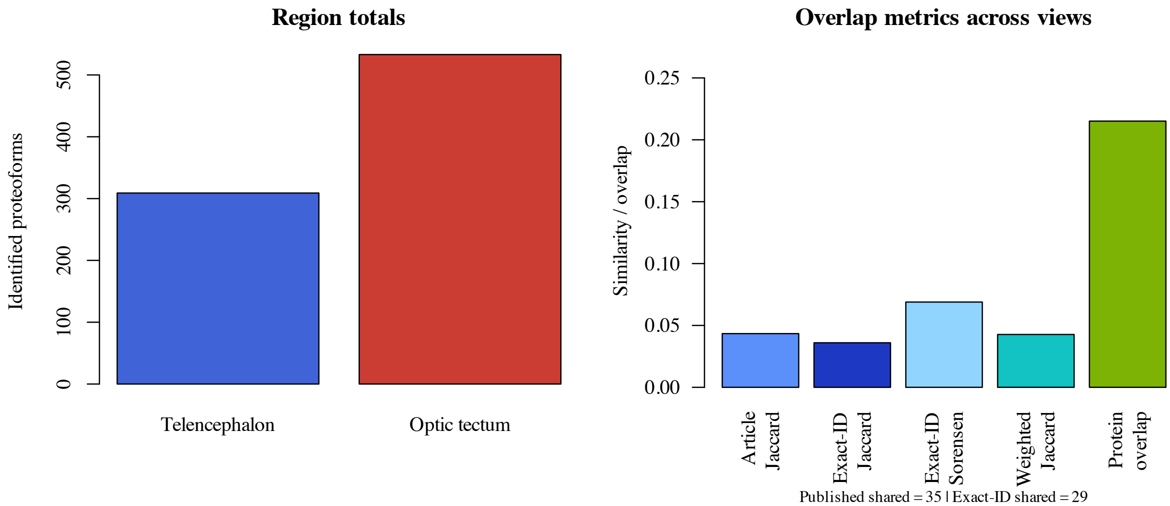

Figure 1: Regional totals and overlap metrics. The deposited region spreadsheets contain 303 exact IDs in telencephalon and 531 in optic tectum, with 29 exact-ID overlaps. This spreadsheet-based count is slightly stricter than the article aggregate, but both support the same low-overlap picture.

Table 3: Axis-level alignment compared with the naive baseline implied by regional prevalence.

Functional axis	Observed	Wilson 95% CI	Baseline	Lift
Telencephalon-associated	0.9200	0.7503–0.9778	0.1875	0.7325
Optic-tectum-associated	0.9903	0.9470–0.9983	0.8125	0.1778

An independent panel curated from the broader zebrafish literature reaches the same qualitative conclusion. Adult zebrafish myelin markers and optic-tectum radial-glia markers yield 32 matched counts and only 3 spillover counts, for an alignment fraction of 0.9143 and a one-sided exact p value of 6.68×10^{-4}. The panel is smaller and composition-oriented, so it does not replace the original functional families. It does show that the regional split is not an artifact of one favored vocabulary.

3.3 The PTM layer is interesting, but the acetylation signal is not yet stable

The PTM layer tells a more cautious story. The original article reported 807 proteoforms in total, 358 with mass shifts and 211 with N-terminal acetylation, so the dataset is plainly proteoform-rich rather than a protein list in another format. Yet the specific acetylation contrast is less secure than the regional overlap and marker results. In the matched marker subsets, 14 of 23 telencephalon-associated proteoforms are N-terminally acetylated, compared with 16 of 102 optic-tectum-associated proteoforms. We therefore treated the PTM follow-up as a sensitivity problem rather than as a discovery claim. In a coarse logistic model with region as the outcome and acetylation, log precursor mass, sequence span length, and first-residue position as predictors, the effect weakens markedly. The adjusted odds ratio falls to 2.94 (95% CI 0.70–12.30; $p = 0.140$), and the first-residue$\leq$ 2 subset shows no enrichment. The safest reading is therefore that acetylation remains a lead for future work rather than a settled regional program.

The duplicate analyses help explain why a small study can nevertheless reveal a clear regional structure. Duplicate runs recovered a mean of 232 proteoforms in telencephalon and 356 in optic tectum, corresponding to recovery fractions of 0.7508 and 0.6680 relative to the regional totals. The design remains limited, but the

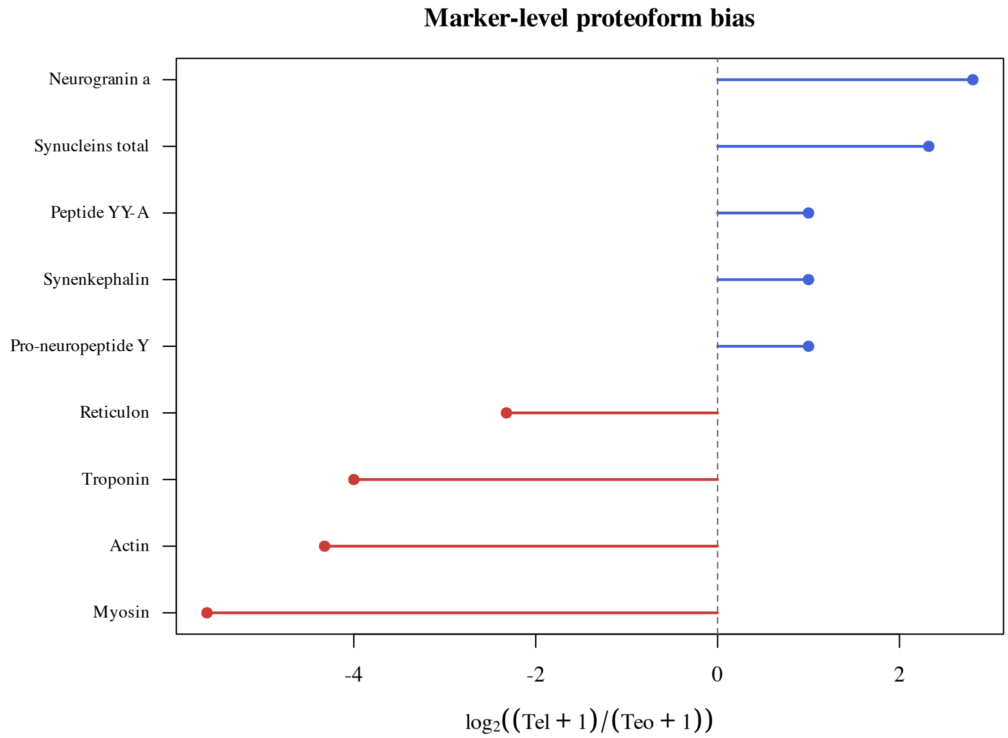

Figure 2: Marker-level proteoform bias across 128 curated proteoform observations in nine marker families. Positive values favor telencephalon and negative values favor optic tectum.

regional split is not a single-run accident.

4 Discussion

The simplest reading of these data is that adult zebrafish telencephalon and optic tectum are organized by different proteoform complements. That conclusion does not rest on one metric or on a handful of chosen examples. It appears in the overlap structure of the deposited spreadsheets, in the distribution of biologically interpretable marker families, and in the failure of conservative sensitivity analyses to erase the signal.

The biology of the split is at least as important as its magnitude. Telencephalon is enriched for neuropeptide and synaptic entries consistent with a forebrain region devoted to integration, learning, and behavioral regulation. Optic tectum is enriched for reticulon-, actin-, myosin-, and troponin-linked proteoforms consistent with retinotectal circuitry and the cellular architecture needed for visually guided orienting. The point is not simply that the two regions differ. It is that the difference remains intelligible when viewed against established regional function.

That regional logic would be easy to blur in a protein-collapsed analysis. Protein overlap is substantially broader than proteoform overlap, which means that much of the distinction lies in the resolved proteoform state rather than in simple protein presence or absence. For a region-resolved nervous system dataset, that is the central biological lesson. The proteoform layer preserves a regional texture that broader summaries dilute.

The independent composition panel supports the same conclusion from a different angle. Myelin markers remain associated with telencephalon, and optic-tectum radial-glia markers remain associated with optic tectum. The panel is small, but it matters because it shows that the result does not disappear when the

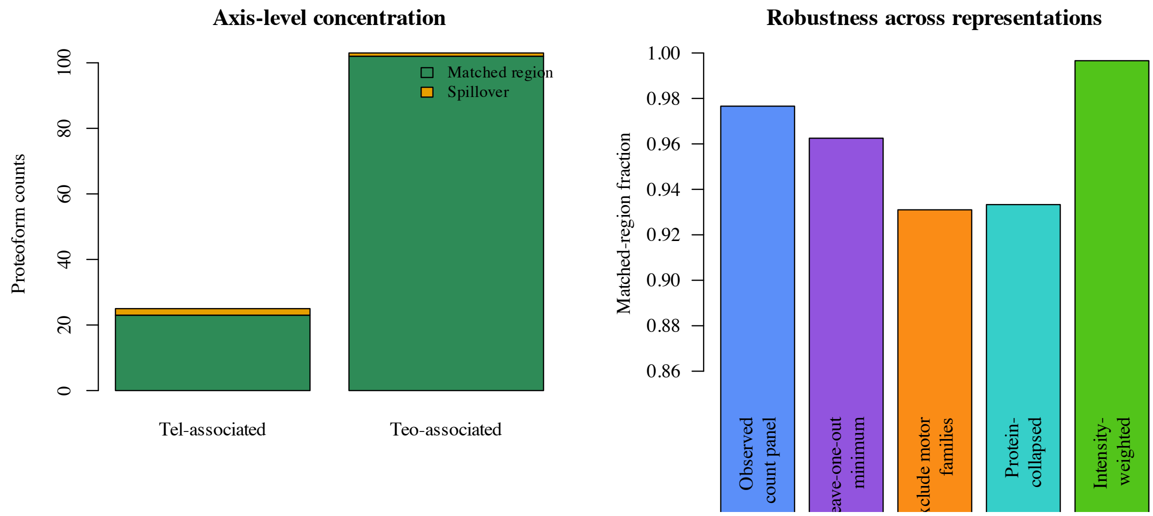

Figure 3: Axis-level concentration and robustness summaries for the nine curated marker families. The overall alignment signal remains high after leave-one-out perturbation, after excluding motor families, after collapsing to proteins, and after weighting by mean duplicate intensity.

Table 4: Sensitivity checks requested by the external review. The motor-family exclusion uses a Haldane correction because one spillover cell is zero after exclusion.

Scenario	Alignment	Odds ratio	One-sided p
Observed panel	0.9766	1173.0	5.12×10^{-22}
Shift 1 matched count per axis	0.9609	370.3	2.01×10^{-19}
Shift 2 matched counts per axis	0.9453	175.0	3.73×10^{-17}
Shift 3 matched counts per axis	0.9297	99.0	3.93×10^{-15}
Exclude myosin, actin, troponin	0.9310	84.6	6.32×10^{-4}

vocabulary is widened beyond the original paper's favored examples.

These findings also help place the study within the broader landscape of spatial top-down proteomics. Imaging-based top-down workflows such as LAESI and nano-DESI MSI bring direct spatial fidelity and on-tissue proteoform mapping (Kiss et al., 2014; Yang et al., 2022). Microdissected separations such as LCM-CZE-MS/MS occupy the complementary position: they trade direct within-section imaging for deeper coverage within anatomically defined regions. The present analysis depends on that tradeoff. It is precisely the added depth of the region-resolved proteoform tables that makes the overlap structure and marker architecture legible.

The deposited spreadsheets also reveal a practical lesson about proteoform analysis. The gap between the article's 35 shared proteoforms and the strict exact-ID count of 29 appears to arise from representation rather than from biology. Relaxed matching changes the count, but not the inference. Standards such as ProForma matter precisely because they help separate notational variation from biological difference (LeDuc et al., 2018, 2022). The present deposit does not contain enough localization metadata for full ProForma 2.0 serialization of every entry, but it contains enough structure to show that the regional conclusion is stable across reasonable canonicalization choices.

The detectability and identifier-error analyses sharpen the same point. Correcting for uneven recovery raises the exact-ID Jaccard only modestly, and conservative error bounds do not drive the two regions toward

Table 5: Technical sensitivity values reported in the original study and extended with exact-ID duplicate-incidence counts from the deposited region spreadsheets.

Region	Mean	SD	CV	Recovery	exact-ID q_1	Chao2
Telencephalon	232	28	0.1207	0.7508	152	379.5
Optic tectum	356	88	0.2472	0.6680	351	873.2

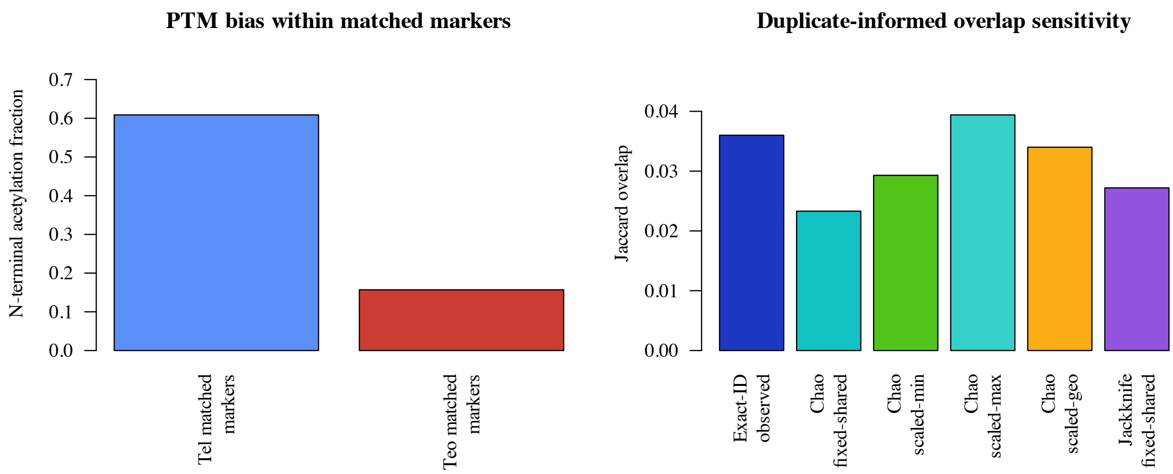

Figure 4: PTM and missingness-sensitive follow-up analyses. Left: N-terminal acetylation in the matched-marker subsets (23 telencephalon counts and 102 optic-tectum counts). Right: duplicate-informed Chao2 sensitivity changes the exact-ID Jaccard modestly, but never enough to erase the regional separation.

a high-overlap case. To overturn the result, one would need far more hidden sharing than the duplicate structure or the published FDR values make plausible.

By contrast, the acetylation result softens as soon as detectability is modeled explicitly. That is a useful outcome, not a disappointment. It marks the point at which the dataset stops warranting a strong biological claim. The strongest statement supported here is therefore about regional proteoform separation, not about a settled regional PTM program.

4.1 Limits of inference

This remains a pilot-scale study built from deposited spreadsheets rather than from a new experiment or a full raw-file reprocessing workflow. The strongest claims are therefore about regional separation and about the biological direction of that separation, not about every proteoform in the adult zebrafish brain. The duplicate-labeled spreadsheets are sufficient for missingness bounds, normalization checks, marker traceability, occupancy-style detectability correction, and a limited PTM screen, but they do not expose the feature histories, localization confidence, or cross-region replicate structure that a raw TopPIC export would provide (Choi and Liu, 2022). Even so, the most obvious technical objections do not overturn the regional signal.

The next steps are clear. The strongest extension would combine the public raw files with processed search intermediates so that the same question could be asked at feature level and under richer missingness models (Karpievitch et al., 2012; Lazar et al., 2016). Additional brain regions would test how general this axis of separation is, and more explicit purity measurements would strengthen the anatomical interpretation. For the present dataset, however, the main point is already secure: adult zebrafish telencephalon and optic tectum are distinguished by different proteoform profiles, and those profiles track known regional biology.

5 Conclusion

Adult zebrafish telencephalon and optic tectum are separated by distinct proteoform profiles, not merely by small shifts in abundance. The proteoforms that distinguish the two regions follow the expected biology of forebrain integration on one side and visuomotor architecture on the other, whereas the apparent acetylation asymmetry remains provisional. Public top-down datasets of this kind can therefore yield genuine neurobiological insight when the proteoform layer is read directly and with appropriate restraint.

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A Reproducibility Appendix

A.1 Local files

The package is backed by the source files in `data/`, the local analysis and figure scripts in `results/`, the build scripts in `code/`, and the environment metadata in `code/environment_versions.json`.

A.2 Formulas

The main derived formulas are the Jaccard overlap, specialization fractions, marker-level stabilized log-bias, Wilson intervals for axis-level alignment fractions, an approximate odds-ratio interval for the 2×2 matched-versus-spillover table, an exact test on the same table, a prevalence-adjusted overlap-significance check based on the fixed-margin hypergeometric null, a two-run detectability correction based on duplicate rediscovery, and a conservative misidentification ceiling based on paired collapse of region-unique IDs.

A.3 Extended computational details

The main text reports only the quantities needed to follow the biology. The full computational path remains local and explicit. Exact-ID overlap is defined on accession-plus-proteoform pairs. The discrepancy diagnostic compares that strict unit with relaxed accession-plus-canonicalized-sequence matches that strip bracketed PTM annotations and punctuation while preserving accession identity. Protein-collapsed summaries merge rows at the protein identifier level. Abundance-weighted comparisons use mean duplicate intensities within each region and report both weighted Jaccard and Bray–Curtis similarity.

For prevalence-adjusted overlap, the observed telencephalon and optic-tectum margins are held fixed and the overlap count is evaluated against the fixed-margin hypergeometric null. The paper reports the observed Jaccard, the marginal-prevalence baseline under independence, the centered Jaccard, and the exact lower-tail p value. For marker alignment, the analysis uses a matched-versus-spillover table together with Wilson intervals, an approximate odds-ratio interval, and an exact directional test. Family-size-preserving randomization keeps family sizes and the global telencephalon/optic-tectum label totals fixed while shuffling labels across the panel counts.

Missingness and abundance sensitivities are likewise kept local and inspectable. The duplicate `Match`/`No match` flags in the deposited spreadsheets provide the incidence counts used for Chao2 and first-order jackknife lower-bound richness calculations. Blank intensity cells are treated as absent detections in the baseline analysis, and the MNAR-style sensitivity fills missing exact IDs to half of the region-specific 5th-percentile or 10th-percentile nonzero intensity. Alternative abundance normalizations include per-run total-intensity scaling before duplicate averaging, region-level total-sum scaling, upper-quartile scaling, and median-ratio scaling on shared exact IDs.

The detectability bound uses the same duplicate structure in a different way. After collapsing each region to unique accession-plus-proteoform IDs, each exact ID is treated as detected in one run or both runs. Under a simple homogeneous two-run model, the duplicate-rediscovery counts give a region-specific per-run detection probability and therefore an estimated "seen at least once" probability for that region. The observed cross-region shared exact IDs are then divided by the product of those two region-level detection probabilities to estimate how much overlap could be hidden by uneven duplicate recovery alone. This is intentionally simpler than a full hierarchical model, but it answers the reviewer's immediate question with quantities that are actually supported by the duplicate-labeled spreadsheets.

The misidentification sensitivity files are equally conservative. They do not claim to estimate the true false-discovery process. Instead, they ask how much the exact-ID Jaccard could rise if an equal fraction of region-unique exact IDs in both regions were actually paired representation or identification errors. The 1% and 5% scenarios track the PrSM- and proteoform-level FDR values reported in the source supplement. A second set of rows asks what error fractions would be required to raise the exact-ID Jaccard to 0.10 or all the way to the observed protein-level overlap.

A.4 Accession conflicts, overlap statistics, and marker traceability

When the same proteoform-like string appeared under more than one accession, the primary overlap analysis retained accession identity rather than collapsing immediately to gene or protein level. The reason is biological and practical: premature collapse can make distinct reported entries appear shared when the deposited spreadsheets still distinguish them. Protein-level summaries are therefore presented as sensitivity analyses, not as the primary unit of inference.

The prevalence-adjusted overlap statistics are reported for all three main representations. Under strict exact-ID matching, the observed Jaccard is 0.0360 against an independence baseline of 0.3152, yielding a centered Jaccard of -0.2792 with $p = 5.56 \times 10^{-183}$. Under canonicalized matching, the observed Jaccard is 0.0615 against an independence baseline of 0.3243, yielding a centered Jaccard of -0.2627 with $p = 5.01 \times 10^{-143}$. After protein collapse, the overlap rises to 0.2151, but still remains far below the protein-level independence baseline of 0.4280 with centered Jaccard -0.2130 and $p = 5.84 \times 10^{-33}$.

The marker-based interpretation is designed to remain externally auditable. The file `results/marker_family_membership.csv` links every curated family member to the source spreadsheet, row number, accession, proteoform string, matching rule, and matched-versus-spillover assignment. The file `data/evidence_traceability.json` mirrors that mapping at the claim level so that each headline statement in the paper can be traced back to a concrete local artifact.

A.5 Scope note

The computation uses values recoverable from the published article and the deposited region spreadsheets. It does not claim a full raw-data reprocessing pipeline.

A.6 Canonicalization and missing-intensity note

Canonicalization is used only as a discrepancy diagnostic, not as the paper's primary biological unit. The strict analysis unit remains accession plus proteoform string. In the diagnostic view, bracketed PTM annotations, punctuation, and non-sequence characters are removed before matching. The file `results/canonicalization_examples.csv` now separates each example into sequence core, modification tokens, residue window, and canonicalized sequence so that the discrepancy can be inspected in a ProForma-oriented way (LeDuc et al., 2018, 2022). We do not emit full ProForma 2.0 strings for the whole dataset because the deposited spreadsheets do not expose the localization metadata needed for exact standard-compliant serialization. Blank intensity cells in the deposited spreadsheets are treated as absent detections in the baseline analysis and contribute zero intensity; the reviewer-driven MNAR sensitivity file records two low-intensity-limit alternatives.

A.7 Reproducibility metadata

The package records software versions and platform metadata in `code/environment_versions.json`. For review, the exact code and derived-table bundle accompanies the manuscript in `code/`, `results/`, and `data/`. The same metadata file records the public project repository URL, but no archival DOI has yet been minted for this specific example package.

A.8 Detectability-adjusted acetylation model

The covariate-adjusted acetylation analysis is intentionally simple. It uses the matched marker rows only. Region is coded as the binary outcome, and the predictors are acetylation status, log precursor mass, sequence span length, and first-residue position, with the continuous covariates standardized before fitting. The goal is not to claim a complete model of proteoform detectability. It is to ask whether the apparent acetylation asymmetry survives once the strongest public-table correlates of detectability are placed in the same coarse model. It does not. A second analysis then restricts the rows to first residue ≤ 2 as a stricter N-terminal geometry screen.

A.9 Public provenance note

The journal supplement records that the deposited spreadsheets were generated with 1% PrSM FDR and 5% proteoform-level FDR in TopPIC and quantified across duplicate runs in TopDiff. The same supplement also embeds `Tel2` and `Teo2` sheets that match the standalone PRIDE XLSX files exactly. The PRIDE landing page lists the four raw duplicate files, but this package does not claim a raw-file reprocessing study.

A.10 Exact rebuild path

The package can be rebuilt locally with:

```
cd examples/codex-zebrafish-brain-paper-14rounds
./code/build_inputs.sh
./code/build_paper.sh
```

The analysis step should regenerate the summary metrics, exact-ID overlap files, marker-membership and robustness files, the run-pair and bootstrap sensitivity files, and the figure assets. The paper step should regenerate `paper/main.pdf`.

A.11 Marker-panel curation rule

The marker panel is intentionally conservative. A family enters the panel only if the source proteomics paper names it in the biological interpretation of telencephalon or optic tectum and if that interpretation can be connected to established zebrafish regional function in the cited background literature. The curated counts and family labels are materialized in `data/curated_evidence.json`, while the corresponding exact source-table members, row numbers, and matching rules appear in the marker-membership CSV and the traceability JSON.

A.12 Why the curation unit is still a marker family

The deposited region spreadsheets do expose reusable machine-readable proteoform IDs, and this version of the package takes advantage of that fact. Even so, the strongest biological interpretation in the paper still runs through the marker families named in the original article rather than through a full proteoform-by-proteoform atlas. That is deliberate. The exact-ID files strengthen traceability and missingness sensitivity, but the functional claim remains aligned with the families that the original paper itself foregrounded as biologically interpretable.

A.13 Leave-one-out and sensitivity files

The full leave-one-out table and the reviewer-driven sensitivity scenarios are saved as `results/leave_one_out_alignment.csv` and `results/sensitivity_scenarios.csv`. The main text and Figure 3 summarize the same robustness story.

A.14 Claim-to-evidence ladder

Table 6: Main claims, supporting evidence, and corresponding limits.

Claim	Evidence	Main limit
Proteoform-level regional separation	Article Jaccard 0.0434, exact-ID Jaccard 0.0360, occupancy-adjusted Jaccard 0.0432, 5% misidentification ceiling 0.0540, protein overlap 0.2151	Public tables, not raw-feature reprocessing
Functional markers align with expected region	Axis alignment, exact test, sign check, leave-one-out table, protein collapse, intensity weighting	Curated marker panel only
PTM layer contains a suggestive but unstable acetylation asymmetry	Unadjusted acetylation 14/23 versus 16/102; detectability and covariate sensitivity files	Public tables do not separate biology cleanly from geometry/coverage bias
Package improves auditability	Local code, figures, manifests, exact marker membership, and evidence traceability	Does not replace the source experiment

A.15 Claim-to-file map

The repository-level package keeps the full claim-to-file map in the local README and machine-readable manifests. In-paper, the important point is simply that every headline result in the main text is backed by a corresponding local CSV/JSON artifact plus the figure that summarizes it.